FILTERED COLIMITS OF SHEAF COHOMOLOGY ON A QCQS ALGEBRAIC STACK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Colimits and global sections on sites. Cohomology on Sites, Lemma sites-cohomology-lemma-colim-global with Remark sites-cohomology-remark-colim-global. Provides the four conditions verified in the proof.
- Quasi-compact stacks have affine smooth presentations. Properties of Stacks, Lemma stacks-properties-lemma-quasi-compact-stack. Condition (1).
- Affine refinement of fppf and etale coverings. Topologies, Lemmas topologies-lemma-fppf-affine and topologies-lemma-etale-affine. Condition (2).
- qqs permanence of fibre products of stacks. Morphisms of Stacks, Lemma stacks-morphisms-lemma-quasi-compact-quasi-separated-permanence. Condition (3): W is quasi-compact.

Source reference. Original reference 0GN3: sources/stacks-chow/tags.txt, source lines 20778–20778. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GN4: sources/stacks-chow/tags.txt, source lines 20779–20779. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04YC: sources/stacks-chow/tags.txt, source lines 6425–6425. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 021P: sources/stacks-chow/tags.txt, source lines 2678–2678. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0218: sources/stacks-chow/tags.txt, source lines 2661–2661. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 075S: sources/stacks-chow/tags.txt, source lines 9141–9141. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be a quasi-compact and quasi-separated algebraic stack. Then*

$$\operatorname{colim}_i H^p(\mathcal{X}, \mathcal{F}_i) \longrightarrow H^p(\mathcal{X}, \operatorname{colim}_i \mathcal{F}_i)$$

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is an isomorphism for every filtered diagram of abelian sheaves on $\mathcal{X}$. The same is true for abelian sheaves on $\mathcal{X}_{étale}$ taking cohomology in the étale topology.

Proof. Let $\tau = fppf$, resp. $\tau = étale$. The lemma follows from Cohomology on Sites, Lemma 0GN3 applied to the site $\mathcal{X}_\tau$. In order to check the assumptions we use Cohomology on Sites, Remark 0GN4. Namely, let $\mathcal{B} \subset \mathrm{Ob}(\mathcal{X}_\tau)$ be the set of objects lying over affine schemes. In other words, an element of $\mathcal{B}$ is a morphism $x : U \to \mathcal{X}$ with U affine. We check each of the conditions (1) – (4) of the remark in turn:

(1) Since $\mathcal{X}$ is quasi-compact, there exists a surjective and smooth morphism $x : U \to \mathcal{X}$ with U affine (Properties of Stacks, Lemma 04YC). Then $h_x^\# \to *$ is a surjective map of sheaves on $\mathcal{X}_\tau$.
(2) Since coverings in $\mathcal{X}_\tau$ are fppf, resp. étale coverings, we see that every covering of $U \in \mathcal{B}$ is refined by a finite affine fppf covering, see Topologies, Lemma 021P, resp. Lemma 0218.
(3) Let $x : U \to \mathcal{X}$ and $x' : U' \to \mathcal{X}$ be in $\mathcal{B}$. The product $h_x^\# \times h_{x'}^\#$ in $Sh(\mathcal{X}_\tau)$ is equal to the sheaf on $\mathcal{X}_\tau$ determined by the algebraic space $W = U \times_{x,\mathcal{X},x'} U'$ over $\mathcal{X}$: for an object $y : V \to \mathcal{X}$ of $\mathcal{X}_\tau$ we have $(h_x^\# \times h_{x'}^\#)(y) = \{f : V \to W \mid y = x \circ \mathrm{pr}_1 \circ f = x' \circ \mathrm{pr}_2 \circ f\}$. The algebraic space W is quasi-compact because $\mathcal{X}$ is quasi-separated, see Morphisms of Stacks, Lemma 075S for example. Hence we can choose an affine scheme U'' and a surjective étale morphism $U'' \to W$. Denote $x'' : U'' \to \mathcal{X}$ the composition of $U'' \to W$ and $W \to \mathcal{X}$. Then $h_{x''}^\# \to h_x^\# \times h_{x'}^\#$ is surjective as desired.
(4) Let $x : U \to \mathcal{X}$ and $x' : U' \to \mathcal{X}$ be in $\mathcal{B}$. Let $a, b : U \to U'$ be a morphism over $\mathcal{X}$, i.e., $a, b : x \to x'$ is a morphism in $\mathcal{X}_\tau$. Then the equalizer of h_a and h_b is represented by the equalizer of $a, b : U \to U'$ which is affine scheme over $\mathcal{X}$ and hence in $\mathcal{B}$.

This finished the proof. □